GSA 2025

Claims-Based Frailty Among Newly Admitted Long-term Care Residents

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Disclosure

Dr. Kim received personal fee from Pfizer, Alosa Health (ended on 12/31/2022), and VillageMD (ended on 12/13/2022) for unrelated work.

This research was conducted independently and there is no conflicts of interest to disclose.

Purpose of the study

Admission to long-term care (LTC) marks a sentinel transition and is often associated with poor outcomes

High 1-year mortality	Hospital readmissions
~35%	34.4%

OBJECTIVE

Identify frail older adults at LTC admission for risk stratification

Claims-based Frailty Index (CFI)

➔ CFI quantifies **deficit-accumulated frailty** using diagnostic, procedural, and durable medical equipment (DME) codes from **Medicare claims**

➔ **Uses administrative data** instead of in-person assessments

Data source

5% sample of **fee-for-service (FFS) Medicare** beneficiaries from 2015 to 2019

Minimum data set (MDS) data to assess baseline characteristics including Activities of Daily Living and Cognitive Function Scale (CFS)

Cohort Creation

First long-term care admissions (1/1/2015 to 12/31/2019)
Require LTC for ≥30 consecutive days within the episode

n = 113,442

Hospice stay within 1 year prior
n = 9,427

< 12 months of continuous FFS before admission
n = 53,852

n = 50,163

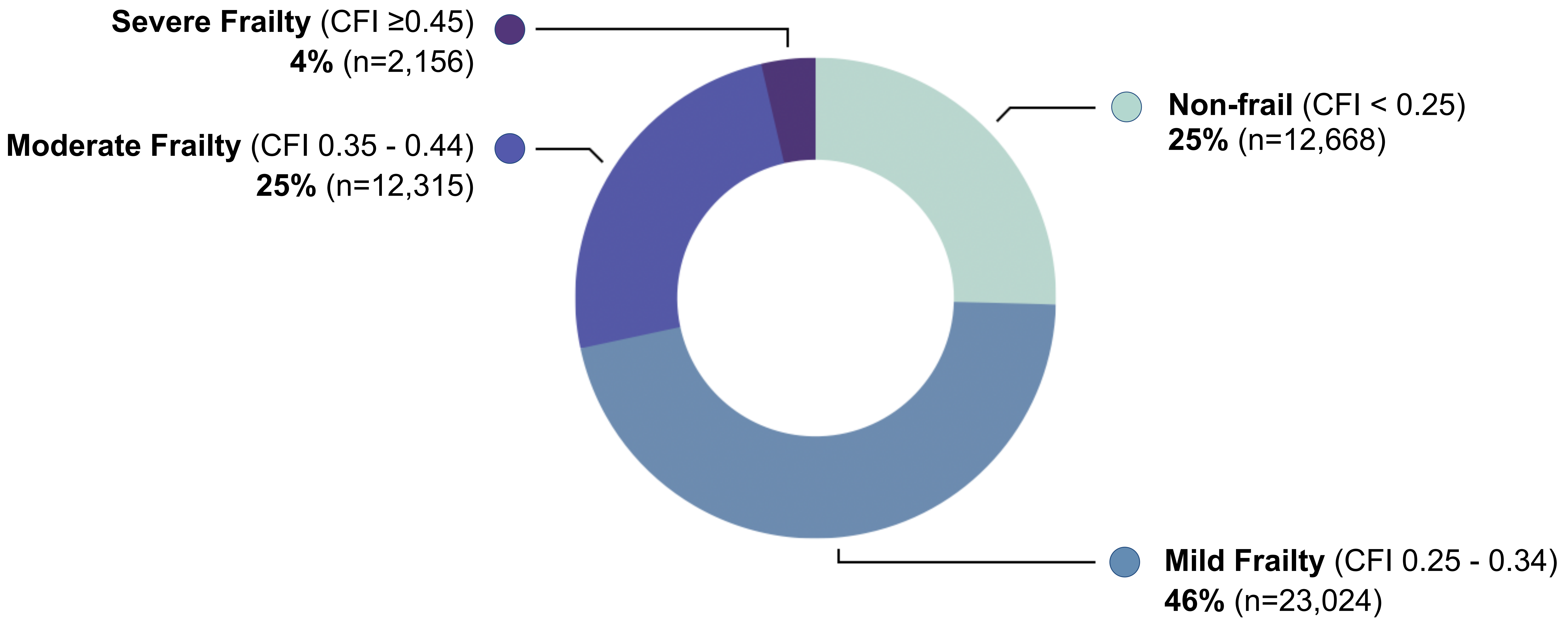
Our primary outcome was mortality

Outcome	Analyses
One year all cause mortality rates and 5 year survival	● Stratification by frailty level, categorical age and sex ● Cox proportional hazards model, adjusting for age, sex, and dual eligibility

Our secondary outcomes were adverse patient outcomes common in the NH population

Outcomes	Analyses
Adverse patient outcomes ○ Hospitalization ○ Emergency department visits ○ Injurious falls ○ Pressure ulcers	Fine-Gray competing-risk models, to account for the competing risk of mortality and administrative censoring
Hospice use among decedents	Median with interquartile range (IQR)

Frailty is common among the LTC population on admission



Baseline characteristics

	Overall n= 50,163	Non-Frail (CFI <0.25) n= 12,668	Mild Frailty (CFI 0.25 - 0.34) n= 23,024	Moderate Frailty (CFI 0.35 - 0.44) n= 12,315	Severe Frailty (CFI ≥0.45) n= 2,156
Age, mean (SD)	**80.4 (12.0)**	**78.2 (13.7)**	**81.0 (11.6)**	**81.4 (10.8)**	**80.4 (10.7)**
Female	**62.7%**	**56.1%**	**62.9%**	**68.2%**	**69.3%**
Race/Ethnicity					
White	**79.7%**	**78.2%**	**80.9%**	**79.9%**	**75.5%**
Black	**11.8%**	**12.5%**	**11.1%**	**11.9%**	**14.7%**
Hispanic	**5.0%**	**5.3%**	**4.5%**	**5.2%**	**7.3%**
Other	**3.5%**	**4.0%**	**3.5%**	**3.0%**	**2.5%**
Dual Eligible	**60.8%**	**54.9%**	**61.1%**	**64.9%**	**69.4%**

Baseline characteristics

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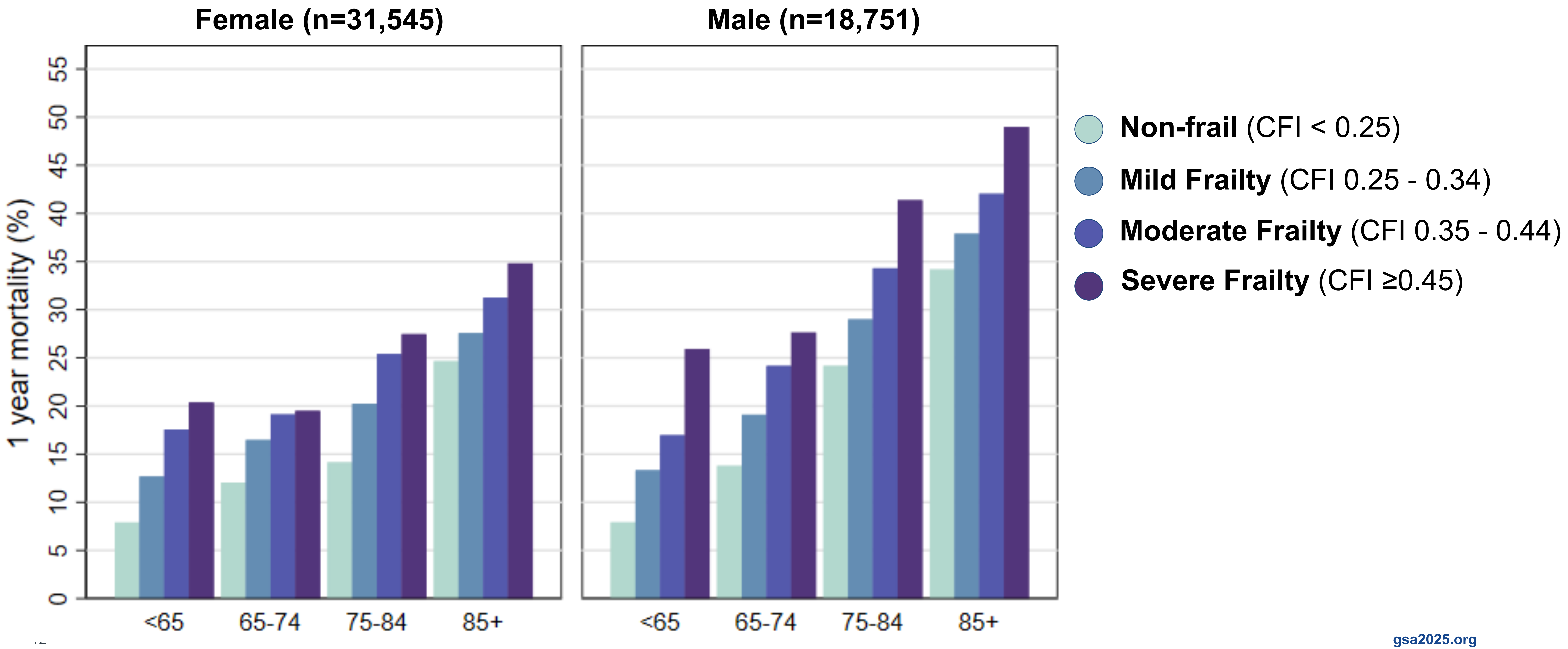
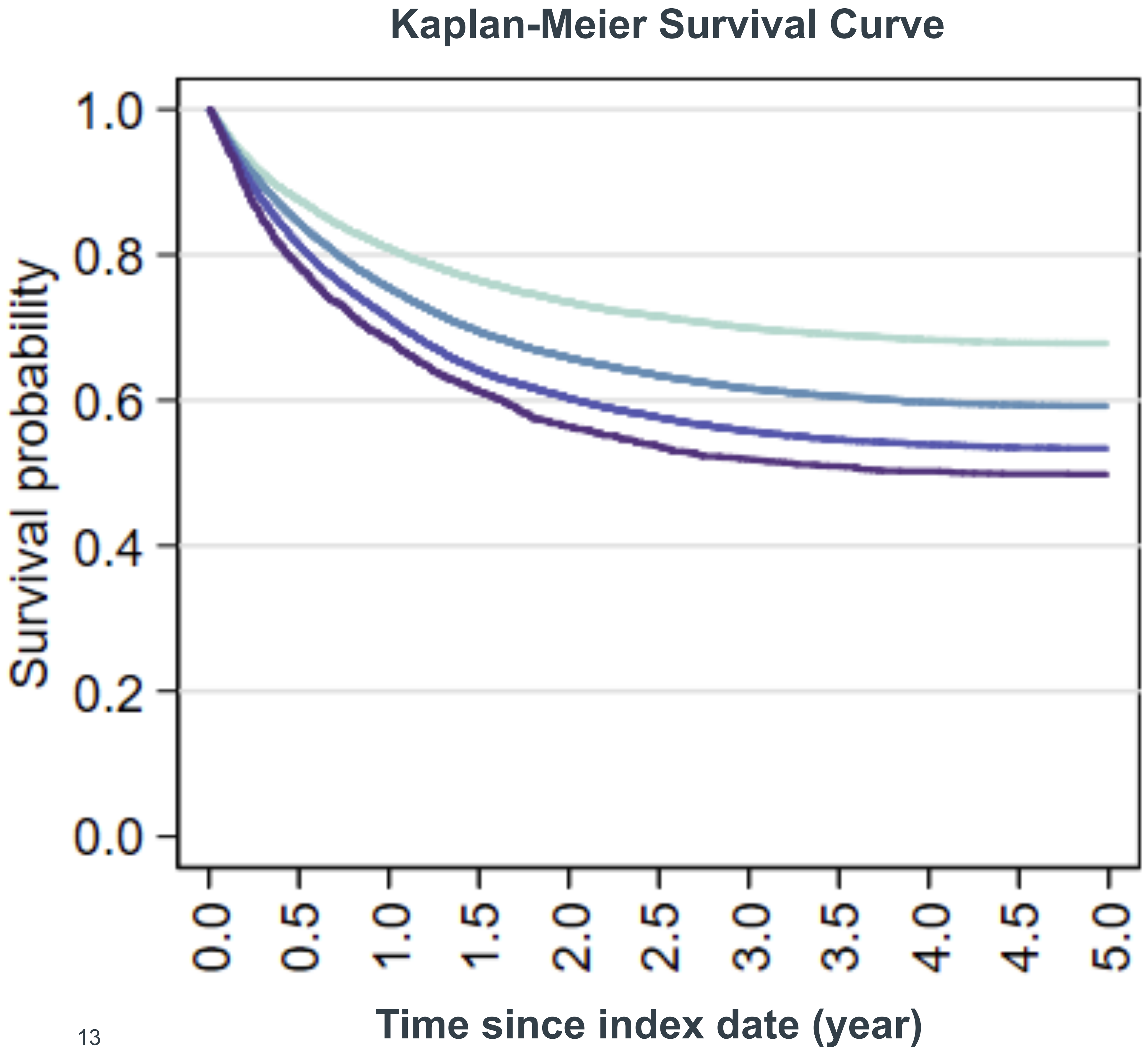
ADL score, mean (SD)	**17.7 (5.0)**	**16.4 (5.6)**	**17.7 (4.9)**	**18.8 (4.4)**	**19.9 (4.1)**
Cognitive Function Scale					
Cognitively Intact	**38.7%**	**49.8%**	**36.1%**	**32.9%**	**35.4%**
Mildly Impaired	**26.1%**	**24.0%**	**26.3%**	**27.6%**	**28.2%**
Moderately Impaired	**31.4%**	**23.3%**	**33.7%**	**35.3%**	**30.5%**
Severely Impaired	**3.8%**	**3.0%**	**3.9%**	**4.2%**	**5.9%**

Note: Cognitive Function Scale (CFS) and ADL were derived from MDS assessments and had missing values (CFS n = 48,066; ADL n = 49,378). The ADL composite score ranges from 0 to 28, with 0–7 = independent or minimal assistance, 8–20 = moderate dependence, and >20 = severe dependence (higher scores indicate greater functional impairment).
Abbreviations: CFI, Claims-based Frailty Index; ADL, Activities of Daily Living; CFS, Cognitive Function Scale; TIA, transient ischemic attack.

The risk of one-year mortality increased with age, and was higher for male and frail

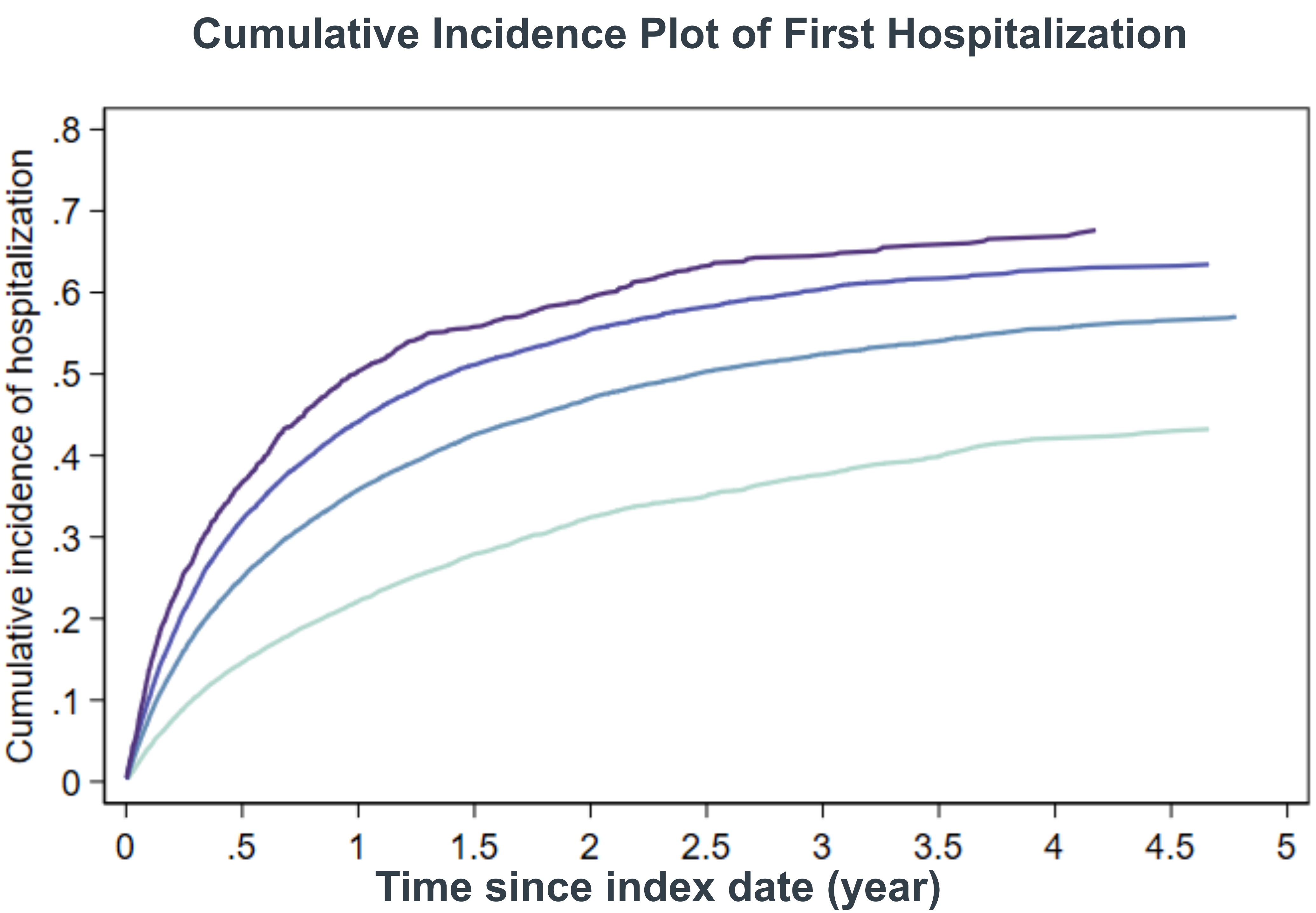


Frailty is associated with higher mortality



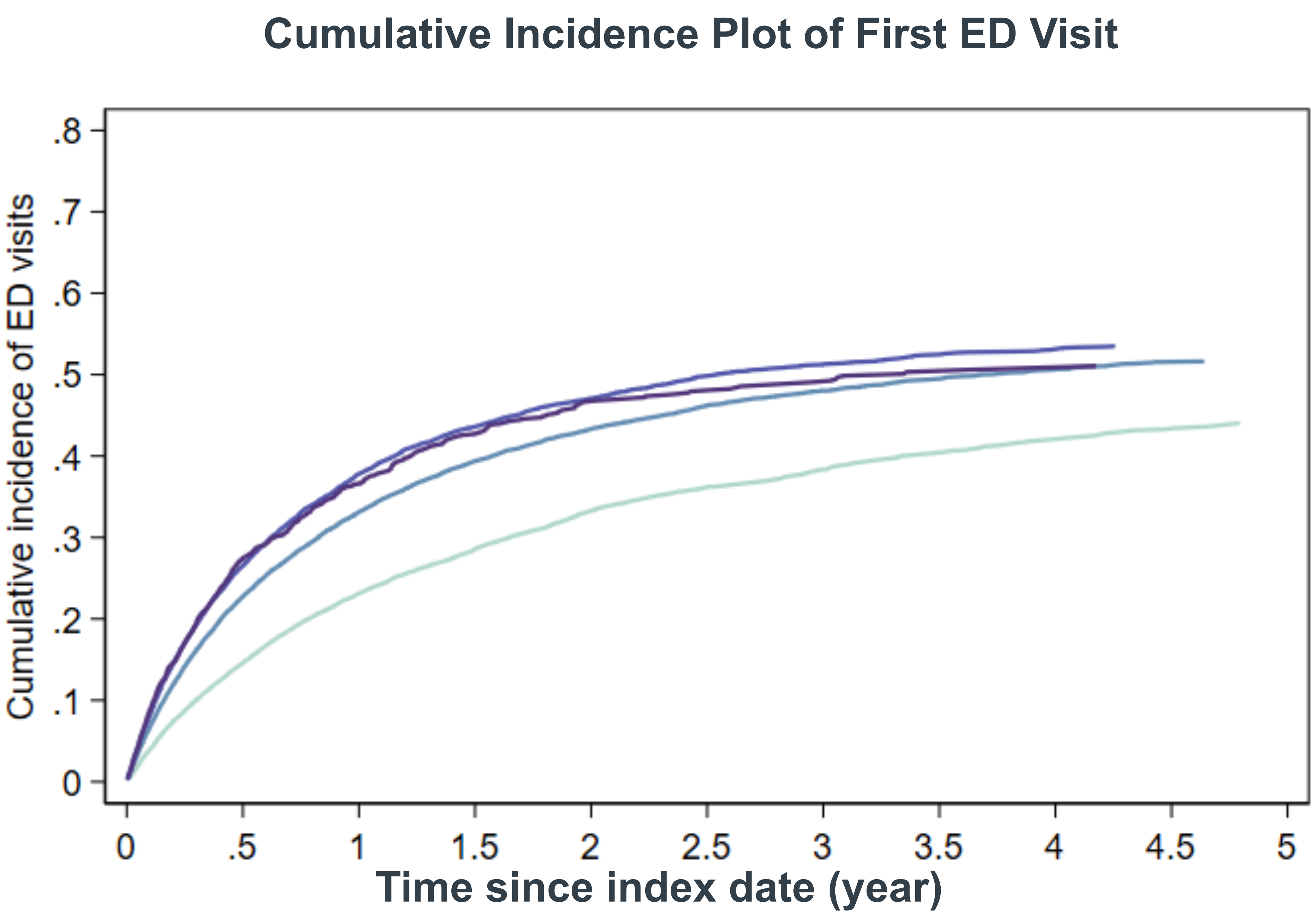
	Adjusted HR [95% CI]
Non-Frail	Ref
Mild Frailty	**1.22** [1.18, 1.27]
Moderate Frailty	**1.45** [1.39, 1.51]
Severe Frailty	**1.62** [1.51, 1.73]

Frailty is associated with earlier hospitalization



	Adjusted HR [95% CI]
Severe Frailty	**2.60** [2.43, 2.79]
Moderate Frailty	**2.26** [2.17, 2.36]
Mild Frailty	**1.72** [1.66, 1.79]
Non-Frail	**Ref**

Frailty is associated with earlier ED visit



	Adjusted HR [95% CI]
Severe Frailty	**1.64** [1.52, 1.77]
Moderate Frailty	**1.72** [1.65, 1.80]
Mild Frailty	**1.50** [1.44, 1.56]
Non-Frail	**Ref**

	Pressure ulcer Adjusted HR [95% CI]	Injurious fall Adjusted HR [95% CI]
Severe Frailty	**1.87** [1.46, 2.39]	**1.11** [0.88, 1.41]
Moderate Frailty	**1.76** [1.54, 2.01]	**1.32** [1.16, 1.51]
Mild Frailty	**1.47** [1.31, 1.66]	**1.28** [1.14, 1.44]
Non-Frail	**Ref**	**Ref**

Hospice use among decedents

	Overall n= 50,163	Non-Frail (CFI <0.25) n=12,668	Mild Frailty (CFI 0.25 - 0.34) n=23,024	Moderate Frailty (CFI 0.35 - 0.44) n=12,315	Severe Frailty (CFI ≥0.45) n=2,156
Death in Hospice (%)	**51.5**	**50.1**	**51.5**	**52.0**	**53.4**
Median days in Hospice among hospice death (IQR)	**16.0 (5.0–67.0)**	**16.0 (5.0–75.0)**	**14.0 (5.0–62.0)**	**14.0 (5.0–55.0)**	**16.0 (5.0–68.0)**

Conclusion

- **Frailty is common** among LTC residents, affecting nearly three-quarters of the population.
- **Frailty levels at admission is associated with increased mortality** and other adverse outcomes.
- Despite this, **hospice care remains underutilized among decedents**, even among the most frail residents, and stays are often short.

Strength and limitation

Strength

- ❖ Addresses a major evidence gap with large-scale data to assess frailty among long-term care residents
- ❖ First measurement of frailty for incident LTC population at admission, enable early risk stratification

Limitation

- ❖ Limited generalizability beyond Medicare FFS
- ❖ Residual confounding

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THANK YOU!